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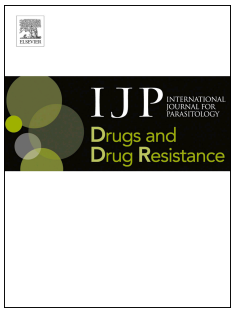

Refugia and anthelmintic resistance: Concepts and challenges

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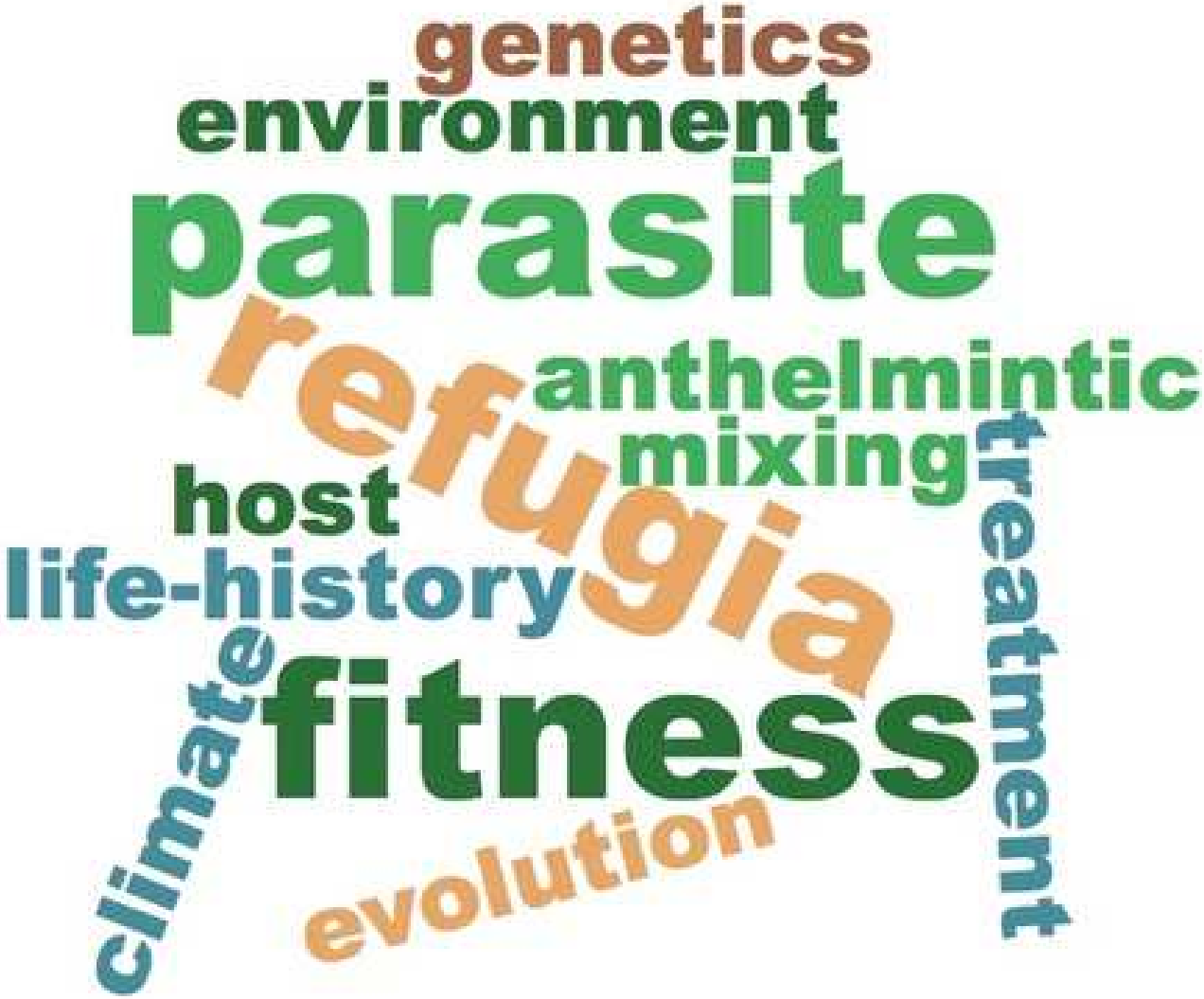
genetics
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refugia
anthelmintic
mixing
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life-history
treatment
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climate
evolution

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1 **Refugia and anthelmintic resistance: concepts and challenges**

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Abstract

Anthelmintic resistance is a threat to global food security. In order to alleviate the selection pressure for resistance and maintain drug efficacy, management strategies increasingly aim to preserve a proportion of the parasite population in 'refugia', unexposed to treatment. While persuasive in its logic, and widely advocated as best practice, evidence for the ability of refugia-based approaches to slow the development of drug resistance in parasitic helminths is currently limited. Moreover, the conditions needed for refugia to work, or how transferable those are between parasite-host systems, are not known. This review, born of an international workshop, seeks to deconstruct the concept of refugia and examine its assumptions and applicability in different situations. We conclude that factors potentially important to refugia, such as the fitness cost of drug resistance, the degree of mixing between parasite sub-populations selected through treatment or not, and the impact of parasite life-history, genetics and environment on the population dynamics of resistance, vary widely between systems. The success of attempts to generate refugia to limit anthelmintic drug resistance are therefore likely to be highly dependent on the system in hand. Additional research is needed on the concept of refugia and the underlying principles for its application across systems, as well as empirical studies within systems that prove and optimise its usefulness.

Key words: refugia, anthelmintic drug, resistance, parasite, control, fitness.

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52 **1. Introduction**

53 The term 'refugium' is classically defined as an area in which a population of
54 organisms can survive through a period of unfavourable conditions. In the context of
55 drug resistance in animal parasites, a refugium refers to untreated hosts or
56 environments that allow the maintenance of drug sensitive parasites in the face of
57 drug exposure. Refugia-based control for livestock parasites has gained increasing
58 traction in the last 20 years, starting with the 'call to arms' by van Wyk (2001), who
59 proposed that refugia should be incorporated more widely into rational
60 anthelmintic use as a means of slowing the spread of resistance. In practice, this
61 frequently relies upon treatment of only a proportion of animals, rather than the
62 whole group, leaving some part of the parasite population untreated and thus free
63 from the selection pressure applied by exposure to drug. At a recent BBSRC-funded
64 meeting in Scotland, an international group of scientists met to discuss the
65 theoretical principles under-pinning refugia-based control strategies and their
66 practical application in the management of drug resistance in parasitic helminths.
67 This short paper summarises the discussion and presents a critical appraisal of the
68 concept of refugia in principle and practice in three host-parasite systems.

69

70 **2. Refugia in helminth parasites - a theoretical perspective**

71 Exploiting refugia as a means of delaying the evolution of resistance has been
72 explored in other biological systems, in particular for the management of
73 agricultural pests (Tabashnik et al., 2008). Here, the planting of insecticidal GM

crops was accompanied by planting of non-GM standard crops to provide refugia for insects. This widespread practice has allowed analysis of the multiple factors determining the success of refugia-based approaches (Jin et al., 2015). These systems have benefitted from adopting a theoretical approach and emphasise that knowledge of both epidemiological and evolutionary dynamics (and their interaction) is essential for the development of optimal control strategies (see for example (Brown and Stankova, 2017). While the use of refugia emerges as a potential control strategy against both *Bacillus thuringiensis* (Bt) insect resistance and anthelmintic resistance, there are important differences. Genetically modified crops producing Bt toxins are ingested by juvenile insect stages (Bravo and Soberon, 2008), whereas advanced larval stages and adult parasites encounter drug product within a host. This alters the time frames, and sequence, over which benefits and costs of resistance play out. In Bt resistance, the benefits of resistance primarily manifest in juvenile survivorship and hence recruitment (Raymond et al., 2011), whereas in helminths a range of effects on larval establishment, adult survival and fecundity may occur.

For gastro-intestinal (GI) nematodes of livestock, control has largely depended upon treatment of all animals in a group with broad-spectrum anthelmintics, which kills susceptible worms, leaving only resistant genotypes to seed the pasture. By minimising exposure to drug, refugia-based control strategies aim to conserve susceptible alleles within the parasite population. The resulting mixture of resistant and susceptible genotypes on pasture should then allow the potential for cross-breeding and/or dilute the frequency of resistant genotypes within a population.

97 Nowadays, best-practice recommendations promote targeted selective treatment
98 (TST), in which only a proportion of a flock is treated (see section 3 for discussion)
99 or strategic timing of drug application to preserve refugia. The success of refugia-
100 based control depends upon many factors; these include the potential fitness costs
101 of harbouring resistant alleles in the absence of drug exposure, the existing level of
102 resistance in the population, the genetic diversity of the parasite population,
103 mechanisms and modes of inheritance of resistance alleles (dominant or recessive),
104 the proportion of the worm population exposed to the drug, which may vary with
105 the particular parasite species (e.g. not all life cycle stages may be exposed or
106 susceptible), and the efficacy and frequency of treatment with a particular drug. In
107 addition, an understanding of environmental factors such as the climate and its
108 effect on the survival of parasite populations in refugia on pasture, and host and
109 pasture management practices, are all relevant for optimising the worm population
110 in refugia. Many of the factors influencing the success of refugia-based approaches
111 in control of GI nematodes were covered in a comprehensive review (Kenyon et al.,
112 2009); in this section we limit our discussion to the impact of the fitness costs of
113 resistance, the influence of parasite life-history traits on the success of refugia-based
114 strategies and the use of modelling approaches.

115

116 **2.1 Fitness costs of resistance**

117 Evolutionary theory predicts that resistance may come at a cost but there are few
118 examples in the literature that have explored fitness costs in metazoan parasites in

119 relation to refugia-based control. The potential fitness costs of drug resistance are
120 central to the discussion of refugia. In a refugium, which is not subject to drug
121 treatment, a fitness cost to a resistance allele will favour and maintain susceptible
122 alleles in a population and so slow the spread of resistance. While theoretically,
123 without a fitness cost, refugia would still slow resistance by reducing selection for
124 resistant alleles, the effect would be less marked than with a fitness cost. *A priori*,
125 one expects such fitness costs. After all, if a protein has been functioning well in a
126 worm for millions of years, ablating or changing it to confer drug resistance is likely
127 to have a cost to the worm in the absence of a drug. Compensatory mutations
128 elsewhere in the genome, however, can ameliorate these costs. In the free-living
129 nematode, *Caenorhabditis elegans*, compensatory mutations can restore wild-type
130 fitness to lines carrying deleterious mutations (Estes et al., 2011). The mechanistic
131 basis of a compensatory mutation has been defined for mercury resistance in
132 *Pseudomonas fluorenscens* (Harrison et al., 2015). In the presence of heavy metals,
133 the plasmid conferring mercury resistance results in significant fitness benefits,
134 while in the absence of heavy metals, the plasmid causes a major disruption to
135 transcription and a large fitness cost. In such populations, secondary, compensatory
136 mutations rapidly emerge that restore the regulation of the transcriptome and,
137 consequently, restore fitness and allow carriage of the plasmid, even in the absence
138 of heavy metals. There is a clear lesson for drug resistance in helminths. Evolution
139 may rapidly find ways to ameliorate any cost of resistance, equalising the fitness of
140 drug resistant and susceptible alleles and so reducing the benefits of refugia in
141 allowing drug susceptible alleles to be maintained. There are few publications that

142 have analysed the fitness costs of anthelmintic resistance; however, a recent review
143 covers the relevant issues in the context of insecticide resistance (Ffrench-Constant
144 and Bass, 2017) and highlights many issues that also apply to anthelmintic
145 resistance, e.g. the need to fully understand fitness costs in field populations of pests.
146 For the insect growth regulator cyromazine, for instance, resistance can be selected
147 rapidly but is highly unstable and efficacy is restored in the absence of selection,
148 implying a high fitness cost (Khan and Akram, 2017). While there are parallels
149 between these respective systems, much remains to be learned about the interplay
150 between the fitness costs of anthelmintic resistance and their possible impact on
151 refugia populations with additional empirical data needed to reach a full
152 understanding.

153

154 **2.2 Parasite life-history traits**

155 The evolution of parasite life history traits (such as establishment, growth and
156 reproduction) will also impact on the development of resistance and refugia. Drug
157 resistance is often considered a question of molecular biology: the variants in drug
158 targets or detoxification systems that are selected for following a drug regime.
159 However, evolution will act on any trait that maximises fitness, and life-history
160 traits of parasitic helminths, unrelated to drug action, have also been proposed to
161 respond to, and ameliorate, the effects of drug (or other) treatment. Although this is
162 not well studied empirically in parasitic helminths, some good examples come from
163 other areas of parasitology. In the rodent malaria parasite *Plasmodium chabaudi,*

there is evidence of selection for greater virulence in response to vaccination with the AMA-1 antigen, but no evidence of selection at the *ama-1* locus (Barclay et al., 2012). Increased virulence of *Plasmodium* results from greater reproduction of blood-stage parasites, which can compensate for the increased level of immunity provided by a vaccination that provides only partial protection. In helminths, Skorping and Read (1998) suggested that migrating stages of helminth species may be less susceptible to drug, with treatment selecting for parasites that migrate for longer and grow larger. Empirical support for this hypothesis came from experimental evolution in *Strongyloides ratti*, where 'fast' and 'slow' lines of worms were generated by passaging eggs obtained from either early or late in an infection, respectively (Paterson and Barber, 2007); drug treatment would be expected to have a similar effect to the 'fast' lines by abbreviating infections and selecting for early reproduction. Consistent with this, 'fast' lines were found to have higher base-line levels of reproduction, but suffered more from the effects of immunity and density-dependence (crowding effects) on reproduction than 'slow' lines. These factors will also play important roles in the success of refugia-based strategies for GI nematodes of livestock, where fecundity levels vary markedly between parasite species, and where the age of the host, and therefore immune status, will influence the numbers of L3 on pasture.

2.3 Modelling refugia

185 In other biological systems where the concept of refugia has been adopted to slow
186 the spread of resistance, mathematical modelling has contributed to understanding
187 the complex interactions between refugia and the prevalence and frequency of
188 resistance (see for example, (Cerda and Wright, 2004; Crowder and Carriere, 2009).
189 In the parasitic helminth field, several approaches to understanding the flow of
190 resistance alleles in nematodes have been published (Leathwick, 2013; Learmount
191 et al., 2016), with recent work explicitly encompassing the impact of refugia on
192 resistance (Park et al., 2015; Cornelius et al., 2016). To address the evolutionary
193 epidemiology of resistance, Park et al. (2015) used S-I (susceptible or infected)
194 models to study two main parameters: the degree of mixing between treated and
195 untreated hosts, and the extent of drug coverage. The expectation was that more
196 mixing would increase the force of infection but also enhance the effect of refugia by
197 increasing dilution by the drug-susceptible strain, limiting the frequency of
198 resistance in treated hosts at a potentially acceptable cost in terms of disease
199 burden. However, the model generated somewhat counter-intuitive results,
200 predicting that the extent of drug coverage and mixing would affect resistance in
201 complex ways, so that increased mixing can either decrease or increase the
202 frequency of resistance depending upon model variables (Park et al., 2015). In
203 particular, for a fixed level of drug coverage, the long-term prediction for the
204 frequency of resistance in treated hosts is minimized at intermediate levels of
205 mixing. Further, for low coverage, increased mixing tends to increase the prevalence
206 of infection in the treated group whereas the opposite is true for high levels of
207 coverage, findings that will be driven by the relative balance of susceptible versus

resistant strains at equilibrium. This model suggested that refugia-based strategies could work, but in a highly conditional way; the apparently counter-intuitive results suggest further investigation is merited into the role density-dependence plays in these findings and what the implications of drug coverage and mixing are for resistance and the prevalence of infection for the whole population (refugia and treated), not just the treated group. Nevertheless, this model highlights the importance of the interaction between mixing and coverage. Variation in mixing may occur by design, such as in a controlled co-grazing strategy (see (Leathwick and Besier, 2014). It may also occur naturally, for example in mass drug administration programmes, such as those in use for soil-transmitted helminths and schistosomiasis, which focus on school-aged children, and thus have the potential to result in age-structured refugia (Webster et al., 2008). These models underscore the complexity of the numerous interactions that affect resistance and thus refugia. Future models could be adapted to include the consequences of recessive or dominant genotypes and polygenic mechanisms of resistance; the role of standing genetic variation versus *de novo* mutation; the parasite life history traits affected by drug treatment (e.g. longevity or fecundity of adult worms) and the potential costs (or benefits) of resistance. Additionally, models may be extended to other ecological transmission settings, particularly those that generate an unequal distribution of parasites in infected hosts such as spatial clumping of free-living parasite stages (Cornell et al., 2003) or vector-borne transmission (Churcher and Basáñez, 2008). Thus, knowledge of both epidemiological and evolutionary dynamics (and their interaction) will be essential to develop the most effective refugia-based strategies.

231

232 **3. Refugia in practice - gastro-intestinal nematodes as a case study**

233 The application of refugia-based strategies to slow the spread of drug resistance in
234 helminths has been best studied in GI nematode infection in small ruminants. While
235 this strategy has gained increasing acceptance in recent years (e.g. see SCOPS,
236 http://www.scops.org.uk), significant knowledge gaps remain. Anthelmintic
237 resistance has a genetic basis and may be either pre-adaptive or spontaneous, such
238 that alleles conferring resistance may either be present in populations prior to drug
239 exposure, or may occur *de novo* during the period of drug exposure (Redman et al.,
240 2015). The frequency of resistance genes should increase as the population is
241 exposed to drugs, but whether the converse is true (i.e. a reversion to susceptibility
242 over time once selection pressure is removed) remains unclear. Multiple studies
243 performed on different continents have repeatedly demonstrated that resistance to
244 the benzimidazole (BZ) class of drugs was not reversible: phenotypic resistance
245 remained high many years after cessation of BZ treatment (Hall et al., 1982; Herd et
246 al., 1984; Borgsteede and Duyn, 1989). In contrast, a 5-year study in New Zealand,
247 where a range of best practices were applied, including refugia-based control and
248 combination anthelmintics, demonstrated partial reversion toward susceptibility for
249 levamisole and ivermectin, but not BZ, in multi-drug resistant *Teladorsagia*
250 *circumcincta* populations (Leathwick et al., 2015). The lack of reversion to
251 susceptibility to BZ compounds may be an indication of longer-term or more
252 frequent exposure of parasites to this drug class and the consequent genetic fixation

253 of resistance in the worm population. It could also indicate differences in the
254 dominance/recessive nature of the alleles conferring resistance to each anthelmintic,
255 or differences in fitness costs.

256 The best characterised of current refugia-based control strategies for GI nematodes
257 is targeted selective treatment (TST) of small ruminants. By treating only a
258 proportion of the flock, most animals and their worms remain unexposed to drug,
259 thereby conserving susceptible alleles. A major challenge with TST, and other
260 refugia-based methods, is deciding which animals to treat. GI nematodes are
261 classically over-dispersed in their host; i.e. a few hosts contain most of the worms
262 (Anderson and May, 1978) and potential indicators to treat grazing ruminants
263 include faecal egg counts, milk yield or various production scores (reviewed in
264 Kenyon et al. (2009); Charlier et al. (2014). However, adoption of a specific TST
265 approach depends on the practicalities of the situation on the farm in question, as
266 any modification of standard practice has to be easy to adopt and cost effective. For
267 example, in *H. contortus* endemic zones, a relatively simple measure of anaemia such
268 as FAMACHA©, in which the colour of the mucus membrane is assessed, has been
269 widely used (van Wyk and Bath, 2002). The feasibility of using the FAMACHA©
270 score to monitor the health of infected goats and maintain refugia in the field was
271 exemplified by studies on smallholder farms in Botswana, where *H. contortus* is
272 endemic. Here a TST approach achieved the same health outcome as treating the
273 whole herd, but for 24% of the drug input (Walker et al., 2015). Interestingly, four
274 years later most of the farmers were still applying the same general approach,
275 demonstrating the potential for sustained impact of a relatively limited intervention

276 in a resource–poor setting. Studies from other regions, such as Brazil, confirm that
277 TST based on FAMACHA© scores can be applied practically by farmers to control
278 parasite burdens while generating refugia (Maia et al., 2015).

279

280 In regions where *T. circumcincta* predominates, lower than expected weight gain
281 has been shown to be a reliable indicator of sheep requiring treatment. For example,
282 in a 5-year study Kenyon et al. (2013) demonstrated that this TST approach could
283 reduce the number of anthelmintic treatments, while maintaining live weight gain
284 and drug efficacy. Furthermore, the TST strategy appeared to conserve the diversity
285 of nematode species present within the host over time, implying a refugium of
286 multiple susceptible species on the farm. In contrast, monthly treatment of the
287 whole flock resulted in the dominance of a single, pathogenic and presumably drug-
288 resistant species, *T. circumcincta* (Melville et al., 2016), a scenario likely to result in
289 clinical disease. Although TST based on weight gain as an index of performance can
290 be semi-automated using electronic identification and weigh crates, further research
291 is needed to develop selection tools for TST that are easy to apply and are
292 economically attractive in different contexts. Further work is also required to
293 ensure that TST strategies use suitable treatment thresholds to preserve production
294 whilst minimising anthelmintic use; if treatment thresholds are too low then more
295 treatments may be administered than necessary, reducing available refugia,
296 whereas thresholds too high could result in production loss. In the future, these
297 could draw on new technologies, e.g. automated monitoring of performance or

behaviour (Vercruysse et al., 2018). It is important to acknowledge that TST could result in increased pasture contamination relative to more conservative whole flock treatment, as more resilient hosts could shed large numbers of eggs while still maintaining productivity (Bisset and Morris, 1996). The higher pasture larval burden, which might be expected from this scenario, although assisting in the maintenance of refugia, might impact production gains in parasite naïve animals (Coop et al., 1982); however, studies to date have shown no negative effect of well-managed refugia on pasture contamination or lamb production (Kenyon et al., 2013).

While TST appears to offer a practical solution to the spread of anthelmintic resistance in some systems, we still have no methods to directly determine the extent to which refugia-based strategies influence the rate of spread of resistance alleles within a worm population. This situation will be remedied as molecular methods are developed for tracking anthelmintic resistance (Laing et al., 2016; Doyle and Cotton, 2019); evidence for the slowing of resistance through refugia will be important if livestock producers are to be persuaded to alter existing management practices in the interests of sustained drug efficacy.

3.1 Environmental considerations in GI nematode studies

In addition to factors that affect the worm population in the host, refugia-based strategies are influenced by environmental factors. Certainly, in temperate climates, the majority of the GI nematode population is free-living on pasture for much of the year. As refugia–based strategies rely on dilution of resistant genotypes with

susceptible genotypes, or breeding between resistant and susceptible worms, the dynamics of L3 on pasture and the optimisation of mixing between genotypes are key considerations. The net consequence of multiple interacting factors on larval availability is hard to predict. Elevated temperatures increase the rate of both development and mortality of GI nematode larvae, leading to non-linear relationships between temperature and transmission potential (Kao et al., 2000). Moreover, migration of L3 from the faecal pat onto the herbage is both strongly moisture-dependent (Wang et al., 2014) and dynamic over short time spans (Rose et al., 2015) and thus pasture-level processes will significantly affect the mixing of susceptible and resistant genotypes and their availability to grazing livestock. Although untreated hosts within a group may generate refugia of predictable size in terms of parasite eggs produced, the size of the refugia will subsequently vary according to climatic and other factors, and may dwindle to insignificant levels or, conversely, become large enough to negatively impact production.

A variety of mathematical models have been used to predict the effect of climate and climate change on the epidemiology of GI nematodes, building upon established frameworks (Smith and Grenfell, 1994). For example, the dynamics of *H. contortus* infection in the UK was studied using simulation models under current and projected climates (Rose et al., 2016). Current studies are elaborating upon these to include genetic mixing between selected populations and refugia. Spatial clumping of L3 is predicted to promote inbreeding (Cornell et al., 2003), so factors

342 determining the fine-scale distribution of L3 in space and time are potentially
343 important considerations for the selection of resistance in GI nematodes, as is the
344 potential of over-lapping generations. The way in which susceptible and resistant
345 genotypes mix is likely to be highly system-specific, suggesting that the generation
346 and effectiveness of refugia will be similarly so.

347

348 While parasite life cycle stages in the environment are typically thought of as a
349 refugium, much may depend upon the drug in use. For example, the macrocyclic
350 lactone (ML), ivermectin, is excreted largely intact in the faeces (Canga et al., 2009)
351 and parasite stages present in the faecal pat may still be exposed to drug in the
352 immediate post-treatment period. Moreover, where parasites actively migrate into
353 the soil they may be exposed to a range of microbial products, e.g. ivermectin itself
354 derives from a soil bacterium *Streptomyces avermitilis* (Campbell, 1985; Him et al.,
355 2009). These data suggest that there may be degrees of refugium in the
356 environment; selective pressure could be applied to free-living nematode stages at
357 genetic loci relevant to drug resistance and thus the assumption that the external
358 environment represents a 'neutral' refugium may be naïve.

359

360 Other environmental factors that can impact refugia include the presence of
361 alternative hosts, which could remove infective larvae and/or deposit new eggs.
362 Wild ungulates, for instance, have been shown both to remove GI nematode L3 from
363 livestock pasture (Walker et al., 2018), and to carry and propagate resistant

genotypes (Chintoan-Uta et al., 2014), and presumably could also import drug-sensitive parasites onto a farm. Different livestock species and age classes on a farm could also be considered alternative hosts, differing from each other in grazing patterns and host competence. Planned rotational grazing could therefore be used to reduce highly resistant L3 populations through removal by refractory hosts, and/or supply susceptible genotypes from young or resilient untreated animals. Practical strategies to manage refugia on grazing land shared by multiple host classes might be feasible, if supported by new research in this area.

4. Refugia in a vector borne helminth - the enigma of dog heartworm

While there is a growing body of work on the use of refugia-based control for GI nematodes, much less is known about the potential of refugia-based control in other host-parasite systems. A case in point is the vector-borne nematode *Dirofilaria immitis* (the dog heartworm). Interestingly, the *D. immitis* scenario appears to contradict much of the dogma surrounding the concepts of refugia, as developed for GI nematodes, in that resistance has developed in a high-refugia situation. A sound grasp of the life cycle and transmission of *D. immitis* is required to explain this finding, along with knowledge of the stages targeted by the MLs used in prophylaxis. MLs were introduced for heartworm control in 1987, and the first proven case of resistance was reported in 2013 (with details subsequently published in 2015, Bourguinat et al. (2015), although the possibility that resistance had emerged was discussed almost ten years previously (Hampshire, 2005). A major cause of the

386 apparent contradiction referred to above relates to the definition of 'resistance' in
387 heartworm versus GI nematodes. For *D. immitis*, resistance is defined at the
388 individual worm and individual animal level while for GI nematodes, resistance is
389 typically defined at the level of the parasite population. Since the MLs are expected
390 to be 100% effective when used prophylactically against *D. immitis* (Seward et al.,
391 1986), any deviation from full activity could constitute resistance. Furthermore,
392 distinguishing resistance from issues of owner compliance is complex (Atkins et al.,
393 2014). MLs target the L3 and L4 stages for the first 4-6 weeks post-infection, during
394 which time the parasite migrates through the tissues of the dog. Once the juveniles
395 arrive in the pulmonary vasculature, around day 90 post-infection, the drug loses its
396 ability to kill the worms (McCall et al., 2001). For *D. immitis*, the population in
397 refugia therefore comprises worms in untreated animals, the adult worm (largely
398 unaffected by the MLs) and the stages in the mosquito (unexposed to MLs). The
399 extent of refugia in *D. immitis* infection is thought to be quite large, since most pet
400 dogs do not receive chemoprophylaxis, and there are large numbers of feral and
401 wild canids (such as coyotes) that are susceptible to infection and that tend to have
402 high infection rates (Prichard, 2005; Brown et al., 2012). In addition, the mosquito
403 populations responsible for transmission can be enormous. According to the
404 accepted dogma, high refugia and low treatment coverage should prevent the
405 development of resistance and consequently many experts thought that resistance
406 would not occur in *D. immitis* (Prichard, 2005). Furthermore, if resistance did
407 appear it was assumed that it would spread rapidly as there is no practical means to
408 prevent it; high societal mobility means that dogs move throughout the country

409 freely and in large numbers, and there is only a single drug class available to prevent
410 the infection.

411

412 However, almost everything we know about resistance in *D. immitis* contradicts
413 such expectations. Resistance is largely confined to the Mississippi Delta region
414 (Pulaski et al., 2014), yet this area has the highest predicted level of refugia for a
415 variety of socio-economic, climatic, physiographic and agricultural reasons. In
416 addition, there appears to be little spread of ML resistant heartworms outside this
417 region. How then to explain such contradictions? It seems that various factors are at
418 play, perhaps the most important being the transmission bottleneck caused by the
419 mosquito intermediate host. The processes involved with infection of mosquitoes,
420 development of the larvae to the infective stage inside the mosquito and
421 transmission to another dog, are complex and can fail at multiple points;
422 consequently, the probability of transmission of resistant worms between dogs via
423 the mosquito appears to be very low. Additionally, there may be a fitness cost
424 associated with resistance, and the impact of any fitness deficit will be exponentially
425 amplified through the multi-step process of the lifecycle. This latter possibility is of
426 particular relevance, as a fitness cost could render the resistant genotype unstable.
427 This would decrease the likelihood of transmission and make it most probable
428 where levels of innate transmission are greatest, such as the Mississippi Delta
429 region. Much remains to be learned about resistance in this particular host-parasite
430 system and factors such as the possible genetic sub-structuring of the population,

431 the level of standing genetic variation, and the mechanism/s of resistance are all
432 important unanswered questions. This case illustrates that risk factors for
433 anthelmintic resistance inferred in one system might not hold in others, and that
434 general principles of how refugia impact on the development of drug resistance are
435 elusive in practice.

436

437 **5. Refugia in a helminth that undergoes clonal expansion**

438 Another host-parasite system for which refugia-based strategies may be applicable
439 is the liver fluke, *Fasciola hepatica*. The *F. hepatica* life cycle involves both a
440 mammalian definitive host, primarily sheep and cattle, and a snail intermediate host.
441 There are some similarities between liver flukes and GI nematodes: e.g. the role of
442 climate on pre-parasitic stages (eggs, miracidia, cercariae and metacercariae), the
443 presence of the parasite in wildlife, the ability of the metacercariae to survive on
444 pasture and their tendency towards spatial aggregation. However, there are several
445 factors specific to fluke infection, such as the ability of adult worms to both self- and
446 cross-fertilize, the clonal expansion within the snail and the availability of snail
447 vectors, all of which will impact potential refugia. Adult parasites have a huge
448 reproductive potential with up to 20,000 eggs produced per fluke per day (Boray,
449 1969; Gonzalez-Lanza et al., 1989) and a single miracidium infecting a snail can give
450 rise to ~3000 metacercariae. Thus a large population exists outside the mammalian
451 host (and therefore in refugia), including a large clonal population in the snail vector.
452 However, clonal expansion also provides the potential for amplification of resistance

genes (Beesley et al., 2017) so the effect of the asexual reproducing stages on refugia is unclear.

Currently several flukicide drugs are licensed to control liver fluke, including triclabendazole (TCBZ), clorsulon, nitroxynil, closantel, albendazole, and oxyclozanide (Coles and Stafford, 2001). Unique amongst these drugs, TCBZ, a benzimidazole derivative, demonstrates high efficacy against both adult and immature fluke (Boray et al., 1983) and is the drug of choice to treat livestock and human infection. In livestock, frequent treatment of whole groups, especially in high-risk years, has been a major driver of drug resistant parasite populations, despite the fact that at the time of treatment there is considered to be multiple refugia: eggs on pasture, viable, un-ingested metacercariae on herbage, parasite stages within infected snails and other farm livestock and wildlife reservoirs. Despite these extensive refugia, resistance to TCBZ was first reported in Australia in 1995, is now found in many locations worldwide and is considered a substantial threat (Kelley et al., 2016; Kamaludeen et al., 2019). As control regimes move away from reliance on TCBZ, resistance to other flukicides is emerging (Novobilsky and Hoglund, 2015). It is clear that liver fluke poses challenges distinct to those relevant for GI nematodes, such as the impact of asexual versus sexual reproduction and direct versus indirect life cycles. Given the high pathogenicity of *F. hepatica*, a strong evidence base is required if refugia-based control strategies, such as leaving a proportion of animals untreated, are to be encouraged.

475 Refugia-based strategies to slow the development of drug resistance in *F. hepatica*
476 have yet to be evaluated, either theoretically or empirically. With this in mind we
477 have the potential to build on an existing mathematical model that describes
478 development of liver fluke within a population of cattle (Turner et al., 2016). This
479 model has been used successfully to investigate the potential value of a prototype
480 vaccine and has since been modified to include the impact of drug treatment.
481 Functions incorporated in the model include host susceptibility (based on parasite
482 aggregation), death rate of immature and mature fluke, fluke fecundity and density
483 dependent effects, and fluke maturation time. The model also includes seasonal
484 addition of metacercariae to the pasture, temperature-dependent mortality of
485 metacercariae on the pasture and consumption of metacercariae by definitive hosts.
486 Currently the model allows investigation of the effect of multiple treatments with
487 anthelmintics of different efficacies in the same year. The timing of each treatment
488 can be specified, as can the proportion of animals treated i.e. determining the size of
489 refugia from infected, untreated hosts.

490

491 As for GI nematodes, in order to derive genetically explicit models that can predict
492 the success of refugia-based strategies, an understanding of the mechanism of
493 resistance, whether resistance is a monogenic or polygenic trait and the mode of
494 inheritance is needed. Ongoing studies are using genetic crosses of drug-resistant
495 and drug-susceptible isolates to map resistance loci (Hodgkinson et al., 2013). To
496 date it is known that a single locus, dominant trait can confer resistance to TCBZ

497 (Hodgkinson et al., unpublished data). However, given the difference between *F.*
498 *hepatica* and nematodes in terms of their biology and transmission, the focus
499 remains on providing empirical data to parameterise appropriate models to
500 determine whether maintenance of refugia could impact the spread of resistance in
501 this system.

502

503 **6. Conclusions**

504 The discussion of three separate host-parasite systems demonstrates that while
505 using refugia to slow the spread of anthelmintic resistance is intuitive, it will likely
506 operate differently under each scenario. To fully understand the specific nuances of
507 refugia-based control, additional studies will be required. Many issues of
508 fundamental significance (see Table 1 for examples) have received little
509 experimental attention and, thus their likely impact on refugia is poorly understood.
510 In a recent article, which canvassed key questions in the livestock helminthology
511 field, 10 of 100 proposed concerned refugia (Morgan et al., 2019). The significant
512 progress in parasitic helminth genomics and a clearer understanding of the
513 mechanisms and evolution of resistance will be important to further our
514 understanding of the applicability of refugia-based strategies to metazoan parasites,
515 and providing appropriate genetic markers with which to measure and best
516 generate and utilize refugia. However, while identifying molecular mechanisms of
517 anthelmintic resistance will give us the tools to properly assess whether refugia can
518 slow the spread of resistance, increased interaction between evolutionary biologists,

519 geneticists, mathematical modellers and parasitologists will be key to maximising
520 the impact of such developments.

521

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528 Table 1: Outstanding questions relating to refugia in practice

Do parasites of different genotypes inter-breed freely?
Is there competition between parasites of different genotype in the host?
Are there always trade-offs in fitness (e.g. growth rate versus fecundity)?
Are fitness costs more or less important than the dilution factor in refugia?
To what extent can refugia work to slow the development of resistance even where fitness costs are absent or minimal?
Are fitness costs of resistance negated by compensatory mutations, and how quickly?
Do experiments with laboratory isolates faithfully mimic what happens in the field?
How much refugia is enough?
Can parasite community replacement be effective as a way of restoring drug susceptibility?

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530 **References**

531 Anderson, R. M., May, R. M. 1978. Regulation and stability of host-parasite
532 population interactions 1 Regulatory processes. J. Anim. Ecol. 47, 219-247.
533 Atkins, C. E. et al. 2014. Heartworm 'lack of effectiveness' claims in the Mississippi
534 delta: computerized analysis of owner compliance--2004-2011. Vet. Parasitol
535 206, 106-113.
536 Barclay, V. C. et al. 2012. The evolutionary consequences of blood-stage vaccination
537 on the rodent malaria Plasmodium chabaudi. PLoS Biolo. 10, e1001368.
538 Beesley, N. J., Williams, D. J., Paterson, S., Hodgkinson, J. 2017. Fasciola hepatica
539 demonstrates high levels of genetic diversity, a lack of population structure
540 and high gene flow: possible implications for drug resistance. Int. J. Parasitol.
541 47, 11-20.
542 Bisset, S. A., Morris ,C. A. 1996. Feasibility and implications of breeding sheep for
543 resilience to nematode challenge. Int. J.Parasitol. 26, 857-868.
544 Boray, J. C. 1969. Experimental fascioliasis in Australia. Advan. Parasitol. 7, 95-210.
545 Boray, J. C. et al. 1983. Treatment of immature and mature Fasciola hepatica
546 infections in sheep with triclabendazole. Vet. Rec. 113, 315-317.
547 Borgsteede, F. H., Duyn, S. P. 1989. Lack of reversion of a benzimidazole resistant
548 strain of Haemonchus contortus after six years of levamisole usage. Research
549 in Vet. Sci. 47, 270-272.
550 Bourguinat, C. et al. 2015. Macrocyclic lactone resistance in Dirofilaria immitis:
551 Failure of heartworm preventives and investigation of genetic markers for
552 resistance. Vet. Parasitol. 210, 167-178.
553 Bravo, A., Soberon, M. 2008. How to cope with insect resistance to Bt toxins? Trends
554 Biotechnol. 26, 573-579.
555 Brown, H. E. et al. 2012. Key factors influencing canine heartworm, Dirofilaria
556 immitis, in the United States. Parasites & vectors 5, 245-254.
557 Brown, J. S., Stankova, K. 2017. Game theory as a conceptual framework for
558 managing insect pests. Curr. Opin. Insect Sci. 21, 26-32.
559 Campbell, W. C. 1985. Ivermectin: an update. Parasitol, Today 1, 10-16.
560 Canga, A. G. et al. 2009. The pharmacokinetics and metabolism of ivermectin in
561 domestic animal species. Vet. Journal 179, 25-37.
562 Cerda, H., Wright, D. J. 2004. Modeling the spatial and temporal location of refugia to
563 manage resistance in Bt transgenic crops. Agric. Ecosyst. & Environ. 102,
564 163-174.
565 Charlier, J. et al. 2014. Practices to optimise gastrointestinal nematode control on
566 sheep, goat and cattle farms in Europe using targeted (selective) treatments.
567 Vet. Record 175, 250-255.
568 Chintoan-Uta, C., Morgan, E. R., Skuce, P. J., Coles, G. C. 2014. Wild deer as potential
569 vectors of anthelmintic-resistant abomasal nematodes between cattle and
570 sheep farms. Proc R. Soc. B-Biol. Sci. 281, 20132985.
571 Churcher, T. S., Basáñez, M. G. 2008. Density dependence and the spread of
572 anthelmintic resistance. Evolution 62, 528-537.

573 Coles, G. C., Stafford, K. A. 2001. Activity of oxyclozanide, nitroxynil, clorsulon and
574 albendazole against adult triclabendazole-resistant Fasciola hepatica. Vet.
575 Record 148, 723-724.
576 Coop, R. L., Sykes A. R., Angus, K. W. 1982. The effect of three levels of intake of
577 Ostertagia Circumcincta Larvae on growth rate, food intake and body
578 composition of growing lambs. J. Agric. Sci. 98, 247-255.
579 Cornelius, M. P., Jacobson, C., Dobson, R., Besier, R. B. 2016. Computer modelling of
580 anthelmintic resistance and worm control outcomes for refugia-based
581 nematode control strategies in Merino ewes in Western Australia. Vet.
582 Parasitol. 220, 59-66.
583 Cornell, S. J., Isham, V. S., Smith, G., Grenfell, B. T. 2003. Spatial parasite transmission,
584 drug resistance, and the spread of rare genes. Proc. Nat. Acad. Sci. USA 100,
585 7401-7405.
586 Crowder, D. W., Carriere, Y. 2009. Comparing the refuge strategy for managing the
587 evolution of insect resistance under different reproductive strategies. J
588 Theoret. Biol. 261, 423-430.
589 Doyle, S. R., Cotton, J. A. 2019. Genome-wide Approaches to Investigate Anthelmintic
590 Resistance. Trends in Parasitol.,35, 289-301.
591 Estes, S., Phillips, P. C., Denver, D. R. 2011. Fitness recovery and compensatory
592 evolution in natural mutant lines of C. elegans. Evolution 65, 2335-2344.
593 Ffrench-Constant, R. H., Bass, C. 2017. Does resistance really carry a fitness cost?
594 Curr. Opin. Insect Sci. 21,39-46.
595 Gonzalez-Lanza, C., Manga-Gonzalez, Y., Del-Pozo-Carnero, P., Hidalgo-Arguello, R.
596 1989. Dynamics of elimination of the eggs of Fasciola hepatica (Trematoda,
597 Digenea) in the faeces of cattle in the Porma Basin, Spain. Vet. Parasitol. 34,
598 35-43.
599 Hall, C. A., Ritchie, L., Kelly, J. D. 1982. Effect of removing anthelmintic selection
600 pressure on the benzimidazole resistance status of Haemonchus contortus
601 and Trichostrongylus colubriformis in sheep. Res. Vet. Sci. 33, 54-57.
602 Hampshire, V. A. 2005. Evaluation of efficacy of heartworm preventive products at
603 the FDA. Vet. Parasitol. 133,191-195.
604 Harrison, E., Guymer, D., Spiers, A. J. , Paterson, S., Brockhurst, M. A. 2015. Parallel
605 compensatory evolution stabilizes plasmids across the parasitism-mutualism
606 continuum. Curr. Biol. 25, 2034-2039.
607 Herd, R. P., Streitel, R. H., McClure, K. E., Parker, C. F.. 1984. Control of hypobiotic and
608 benzimidazole-resistant nematodes of sheep. J. Am. Vet. Med. Assoc. 184,
609 726-730.
610 Him, N. A., Gillan, V., Emes, R. D., Maitland, K., Devaney, E. 2009. Hsp-90 and the
611 biology of nematodes. BMC Evol. Biol. 9, 254.
612 Hodgkinson, J., Cwiklinski, K., Beesley, N. J., Paterson, S., Williams, D. J. 2013.
613 Identification of putative markers of triclabendazole resistance by a genome-
614 wide analysis of genetically recombinant Fasciola hepatica. Parasitol 140,
615 1523-1533.
616 Jin, L. et al. 2015. Large-scale test of the natural refuge strategy for delaying insect
617 resistance to transgenic Bt crops. Nat. Biotech. 33, 169-174.

618 Kamaludeen, J. et al. 2019. Lack of efficacy of triclabendazole against Fasciola
619 hepatica is present on sheep farms in three regions of England, and Wales.
620 Vet. Rec. 184, 502.
621 Kao, R. R., Leathwick, D. M., Roberts, M. G., Sutherland, I. A. 2000. Nematode
622 parasites of sheep: a survey of epidemiological parameters and their
623 application in a simple model. Parasitol. 121, 85-103.
624 Kelley, J. M. et al. 2016. Current Threat of Triclabendazole Resistance in Fasciola
625 hepatica. Trends in Parasitol. 32, 458-469.
626 Kenyon, F. et al. 2009. The role of targeted selective treatments in the development
627 of refugia-based approaches to the control of gastrointestinal nematodes of
628 small ruminants. Vet. Parasitol. 164, 3-11.
629 Kenyon, F. et al. 2013. A comparative study of the effects of four treatment regimes
630 on ivermectin efficacy, body weight and pasture contamination in lambs
631 naturally infected with gastrointestinal nematodes in Scotland. Int. J.
632 Parasitolol. Drugs & Drug Resist. 3, 77-84.
633 Khan, H. A. A., Akram, W. 2017. Cyromazine resistance in a field strain of house flies,
634 Musca domestica L.: Resistance risk assessment and bio-chemical mechanism.
635 Chemosphere 167, 308-313.
636 Laing, R. et al. 2016. Analysis of putative resistance gene loci in UK field populations
637 of Haemonchus contortus after 6 years of macrocyclic lactone use. Int. J.
638 Parasitol. 46, 621-630.
639 Learmount, J., Stephens, N., Boughtflower, V., Barrecheguren, A., Rickell, K. 2016. The
640 development of anthelmintic resistance with best practice control of
641 nematodes on commercial sheep farms in the UK. Vet. Parasitol. 229, 9-14.
642 Leathwick, D. M. 2013. Managing anthelmintic resistance--parasite fitness, drug use
643 strategy and the potential for reversion towards susceptibility. Vet. Parasitol.
644 198, 145-153.
645 Leathwick, D. M., Besier, R. B. 2014. The management of anthelmintic resistance in
646 grazing ruminants in Australasia--strategies and experiences. Vet. Parasitol.
647 204, 44-54.
648 Leathwick, D. M., Ganesh, S., Waghorn, T. S. 2015. Evidence for reversion towards
649 anthelmintic susceptibility in Teladorsagia circumcincta in response to
650 resistance management programmes. Int. J. Parasitol. Drugs & Drug Resist. 5,
651 9-15.
652 Maia, D., Rosalinski-Moraes, F., de Torres-Acosta, J. F., Cintra, M. C., Sotomaior, C. S.
653 2015. FAMACHA(c) system assessment by previously trained sheep and goat
654 farmers in Brazil. Vet. Parasitol. 209, 202-209.
655 McCall, J. W. et al. 2001. Further evidence of clinical prophylactic, retroactive (reach-
656 back) and adulticidal activity of monthly administrations of ivermectin
657 (Heartgard PlusTM) in dogs experimentally infected with heartworms. In: R.
658 L. Seward and D. H. Knight (eds.) Recent Adv. Heartworm Disease:
659 Symposium 01, San Antonio, Texas, USA, 20-22 April, 2001. p 189-200.
660 American Heartworm Society, Batavia.
661 Melville, L. A. et al. 2016. Effect of anthelmintic treatment strategy on strongylid
662 nematode species composition in grazing lambs in Scotland. Parasites &
663 vectors 9, 199-210.

664 Morgan, E. R. et al. 2019. 100 Questions in Livestock Helminthology Research.
665 Trends in Parasitol. 35, 52-71.
666 Novobilsky, A., Hoglund, J. 2015. First report of closantel treatment failure against
667 Fasciola hepatica in cattle. Int. J. Parasitol. Drugs & Drug Resist. 5, 172-177.
668 Park, A. W., Haven, J., Kaplan, R., Gandon, S. 2015. Refugia and the evolutionary
669 epidemiology of drug resistance. Biol. Lett. 11.
670 Paterson, S., Barber, R. 2007. Experimental evolution of parasite life-history traits in
671 Strongyloides ratti (Nematoda). Proc. Biol. Sci. 274, 1467-1474.
672 Prichard, R. K. 2005. Is anthelmintic resistance a concern for heartworm control?
673 What can we learn from the human filariasis control programs? Vet. Parasitol.
674 133, 243-253.
675 Pulaski, C. N. et al. 2014. Establishment of macrocyclic lactone resistant Dirofilaria
676 immitis isolates in experimentally infected laboratory dogs. Parasites &
677 Vectors 7, 494.
678 Raymond, B., Wright, D. J., Bonsall, M. B. 2011. Effects of host plant and genetic
679 background on the fitness costs of resistance to Bacillus thuringiensis.
680 Heredity (Edinb) 106, 281-288.
681 Redman, E. et al. 2015. The emergence of resistance to the benzimidazole
682 anthlemintics in parasitic nematodes of livestock is characterised by multiple
683 independent hard and soft selective sweeps. PLoS Negl. Trop. Dis. 9,
684 e0003494.
685 Rose, H. et al. 2016. Climate-driven changes to the spatio-temporal distribution of
686 the parasitic nematode, Haemonchus contortus, in sheep in Europe. Global
687 Change Biol. 22, 1271-1285.
688 Rose, H. et al. 2015. GLOWORM-FL: a simulation model of the effects of climate and
689 climate change on the free-living stages of gastro-intestinal nematode
690 parasites of ruminants. Ecol. Modelling 297, 232-245.
691 Seward, R. L., Brokken, E. S., Plue, R. E. 1986. Ivermectin vs Heartworm - A status
692 update. Proceedings of the Heartworm Symposium '86, New Orleans,
693 Louisiana, March 21-23, 1986. 1-8.
694 Skorping, A., Read, A. F. 1998. Drugs and parasites: global experiments in life history
695 evolution? Ecol Letters 1, 10-12.
696 Smith, G., Grenfell, B. T. 1994. Modelling of parasite populations: gastrointestinal
697 nematode models. Vet. Parasitol. 54,127-143.
698 Tabashnik, B. E., Gassmann, A. J., Crowder, D. W., Carriere, Y. 2008. Insect resistance
699 to Bt crops: evidence versus theory. Nat. Biotechnol. 26, 199-202.
700 Turner, J. et al. 2016. A model to assess the efficacy of vaccines for control of liver
701 fluke infection. Sci. Rep. 6, 23345.
702 van Wyk, J. A. 2001. Refugia-overlooked as perhaps the most potent factor
703 concerning the development of anthelmintic resistance. Onderstepoort J. Vet.
704 Res. 68, 55-67.
705 van Wyk, J. A., Bath, G. F. 2002. The FAMACHA system for managing haemonchosis in
706 sheep and goats by clinically identifying individual animals for treatment. Vet.
707 Res. 33, 509-529.
708 Vercruysse, J. et al. 2018. Control of helminth ruminant infections by 2030. Parasitol.
709 145, 1655-1664.

710 Walker, J. G., Evans, K. E., Rose Vineer, H., van Wyk, J. A., Morgan, E. R. 2018.
711 Prediction and attenuation of seasonal spillover of parasites between wild
712 and domestic ungulates in an arid mixed-use system. J. Appl. Ecol. 55, 1976-
713 1986.
714 Walker, J. G. et al. 2015. Mixed methods evaluation of targeted selective anthelmintic
715 treatment by resource-poor smallholder goat farmers in Botswana. Vet.
716 Parasitol. 214, 80-88.
717 Wang, T., van Wyk, J. A., Morrison, A., Morgan, E. R. 2014. Moisture requirements for
718 the migration of Haemonchus contortus third stage larvae out of faeces. Vet.
719 Parasitol. 204, 258-264.
720 Webster, J. P., Gower, C. M., Norton, A. J. 2008. Evolutionary concepts in predicting
721 and evaluating the impact of mass chemotherapy schistosomiasis control
722 programmes on parasites and their hosts. Evol. Appl. 1, 66-83.

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Highlights

- **Refugia is currently recommended for strategic control of AR in GI nematodes of livestock**
- **Conditions required for refugia-based control may differ between parasite systems**
- **An improved understanding of the parameters influencing refugia-based control is required**

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